

JAIME GINE PARAISO

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EU citizen · Right to work in the UK - Graduate visa, valid until December 2028

SUMMARY

Final year MSc student in High Performance Graphics and Games Engineering at the University of Leeds, predicted to graduate with First Class Honours in September 2026, with a focus on real-time rendering, game engine development, and high-performance C/C++ programming. Seeking a graphics or engine programming role in the games industry.

PROJECTS

Vulkan Renderer (C++, GLSL) - University coursework

- Implemented a real-time Vulkan renderer using Vulkan 1.4 features.
- Uses a PBR shading model with toggleable post-processing effects such as mosaic.
- Added multiple debug modes, to visualise mip-map level usage, depth and partial derivatives.

Physics Engine in Custom ECS Game Engine (C++) - University Coursework / Hobby Project

- Implemented a fully featured rigid-body physics engine integrated into a custom ECS game engine.
- Includes AABB broad-phase collision detection and narrow-phase full implementations of the GJK and EPA algorithms.
- Supports ray-casting, continuous collision detection via shape-casting, and shape-intersection queries.

Path-Tracing Renderer (C++) - University coursework

- Fully featured CPU Monte Carlo path-tracer using physically correct Blinn-Phong shading and anti-aliasing.
- Supports reflections, refractions with Beer's law, and the Fresnel effect.
- Uses Next-Event Estimation for indirect lighting, enabling caustic formation.

Mesh Processing Toolkit (C++) – University coursework

- Implemented mesh processing utilities with support for indexed face and directed edge formats.
- Built automated mesh repair and simplification tools using a greedy algorithm with Gaussian curvature.
- Developed manifold-ness tests identifying non-manifold edges and vertices, with genus calculation via the Euler formula.

EDUCATION

University of Leeds · September 2025 - September 2026

MSc, High Performance Graphics and Games Engineering

Relevant Modules: Foundation of Computer Graphics, Advanced Rendering, Modelling and Animation, Research Seminar

Universidad Complutense de Madrid · September 2020 - September 2024

Bachelor's Degree, Computer Engineering (Grado en Ingeniería de Computadores)

Awarded Best Academic Record of 2023-2024

WORK EXPERIENCE

Junior Programmer - Inycom · November 2024 - June 2025

- Contributed as a full-stack engineer during planning and development in a large-scale collaborative project.
- Developed a user management REST backend using Spring Boot.
- Built the frontend interfaces for user login and profile management using React.
- Directed and reviewed the work of 3 junior interns, assigning tasks and providing guidance throughout the development cycle.

SKILLS AND INTERESTS

Languages: C++, C, GLSL, C#, Python

Graphics APIs: Vulkan, OpenGL

Tools: Git, Premake, CMake, Renderdoc, Azure DevOps

Engines: Unity, Unreal Engine